

Synoptic CB: Porewater SO₄/Cl

2022-2025 Samples

2025-12-17

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0.1 Grab 2022 and 2023 All Data Files

```
#read in 2022 Dionex Data:
dat22 <- read.csv("2022/COMPASS_SynopticCB_SO4_Cl_2022.csv")
# head(dat22)
colnames(dat22)

## [1] "X"           "Project"      "Region"       "Site"
## [5] "Zone"        "Replicate"    "Depth_cm"     "Sample_ID"
## [9] "Year"        "Month"        "Day"          "Time"
## [13] "Time_Zone"   "SO4_ppm"      "SO4_mM"       "SO4_Conc_flag"
## [17] "SO4_QAQC_flag" "Cl_ppm"      "Cl_mM"        "Cl_Conc_flag"
## [21] "Cl_QAQC_flag" "salinity"    "Analysis_rundate" "Run_notes"
## [25] "Field_notes"

#read in 2023 Dionex Data:
dat23 <- read.csv("2023/COMPASS_SynopticCB_SO4_Cl_2023.csv")
# head(dat23)
colnames(dat23)

## [1] "X"           "Project"      "Region"       "Site"
## [5] "Zone"        "Replicate"    "Depth_cm"     "Sample_ID"
## [9] "Year"        "Month"        "Day"          "Time"
## [13] "Time_Zone"   "SO4_ppm"      "SO4_mM"       "SO4_Conc_flag"
## [17] "SO4_QAQC_flag" "Cl_ppm"      "Cl_mM"        "Cl_Conc_flag"
## [21] "Cl_QAQC_flag" "salinity"    "Analysis_rundate" "Run_notes"
## [25] "Field_notes"
```

```
#read in 2024 Dionex Data:
```

```
dat24 <- read.csv("2024/COMPASS_SynopticCB_S04_Cl_2024.csv")
```

```
# head(dat24)
```

```
colnames(dat24)
```

```
## [1] "X"           "Project"      "Region"       "Site"
## [5] "Zone"        "Replicate"    "Depth_cm"     "Sample_ID"
## [9] "Year"        "Month"        "Day"          "Time"
## [13] "Time_Zone"   "S04_ppm"      "S04_Conc_mM"  "S04_Conc_flag"
## [17] "S04_QAQC_flag" "Cl_ppm"      "Cl_Conc_mM"   "Cl_Conc_flag"
## [21] "Cl_QAQC_flag" "salinity"    "Analysis_rundate" "Run_notes"
## [25] "Field_notes"
```

```
#read in 2025 Dionex Data:
```

```
dat25 <- read.csv("2025/COMPASS_SynopticCB_S04_Cl_2025.csv")
```

```
# head(dat25)
```

```
colnames(dat25)
```

```
## [1] "X"           "Project"      "Region"       "Site"
## [5] "Zone"        "Replicate"    "Depth_cm"     "Sample_ID"
## [9] "Year"        "Month"        "Day"          "Time"
## [13] "Time_Zone"   "S04_ppm"      "S04_Conc_mM"  "S04_Conc_flag"
## [17] "S04_QAQC_flag" "Cl_ppm"      "Cl_Conc_mM"   "Cl_Conc_flag"
## [21] "Cl_QAQC_flag" "salinity"    "Analysis_rundate" "Run_notes"
## [25] "Field_notes"
```

```
all_dat <- bind_rows(dat22, dat23, dat24, dat25)
```

```
all_dat <- all_dat %>%
```

```
  dplyr::select(
```

```
    Project, Region, Sample_ID, Year, Month, Day, Time, Time_Zone,
```

```
    Site, Zone, Replicate, Depth_cm,
```

```
    S04_ppm, S04_Conc_mM, S04_Conc_flag, S04_QAQC_flag, Cl_ppm, Cl_Conc_mM, Cl_Conc_flag, Cl_QAQC_flag,
```

```
    Analysis_rundate, Run_notes, Field_notes
```

```
    # list columns in the order you want them
```

```
  )
```

```
##Set 4 decimal points max
```

```
all_dat <- all_dat %>%
```

```
  mutate(across(where(is.numeric), round, digits = 4)) ##This does make all salinities 0 when Cl_ppm is 0
```

```
## Warning: There was 1 warning in 'mutate()'.
```

```
## i In argument: 'across(where(is.numeric), round, digits = 4)'.
```

```
## Caused by warning:
```

```
## ! The '...' argument of 'across()' is deprecated as of dplyr 1.1.0.
```

```
## Supply arguments directly to '.fns' through an anonymous function instead.
```

```
##
```

```
## # Previously
```

```
## across(a:b, mean, na.rm = TRUE)
```

```
##
```

```
## # Now
```

```
## across(a:b, \(x) mean(x, na.rm = TRUE))
```

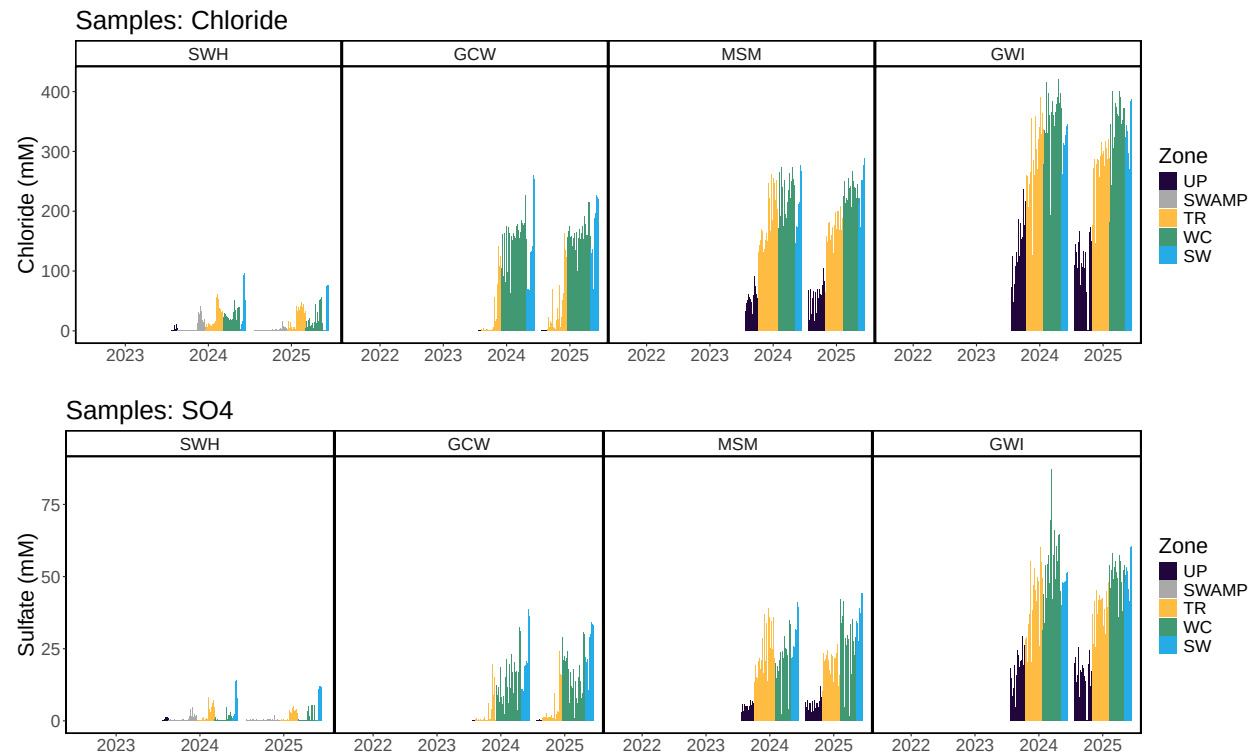
```
##Make Relevant Metadata Sheet
```

0.2 Write out files

```
#write out a csv of all the data to the main folder:
write.csv(all_dat, "COMPASS_SynopticCB_Dionex_AllData.csv")

#write out a csv of the metadata associated with the data:
write.csv(metadata, "COMPASS_SynopticCB_Dionex_Metadata.csv")
```

0.3 Visualize Data by Plot

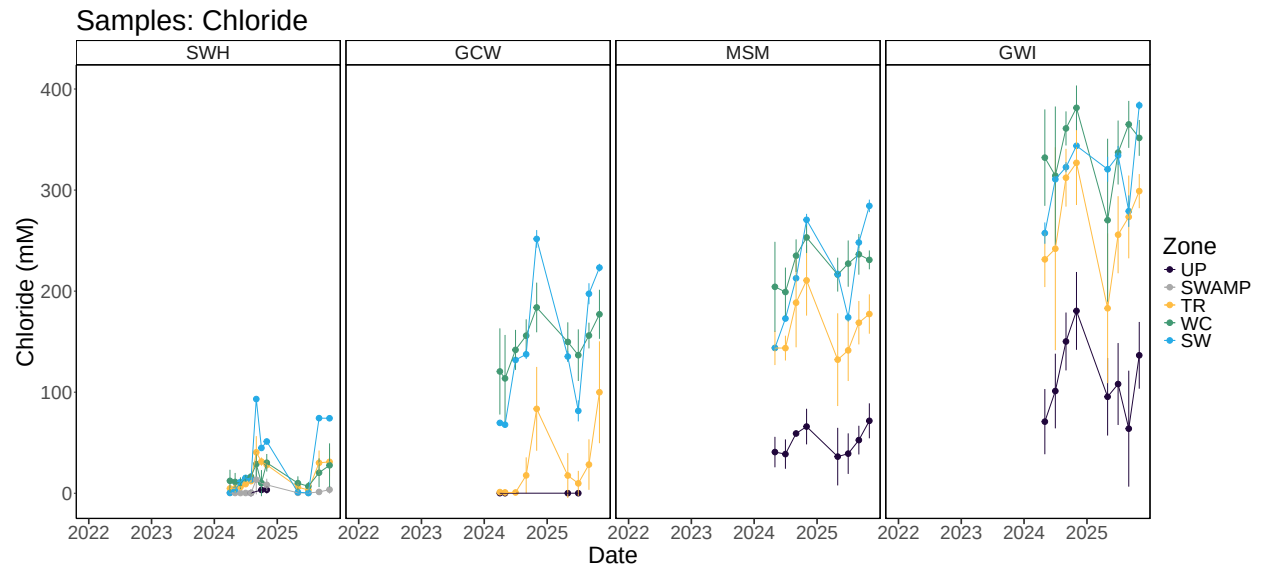


0.4 Summarized data for Site and Zone

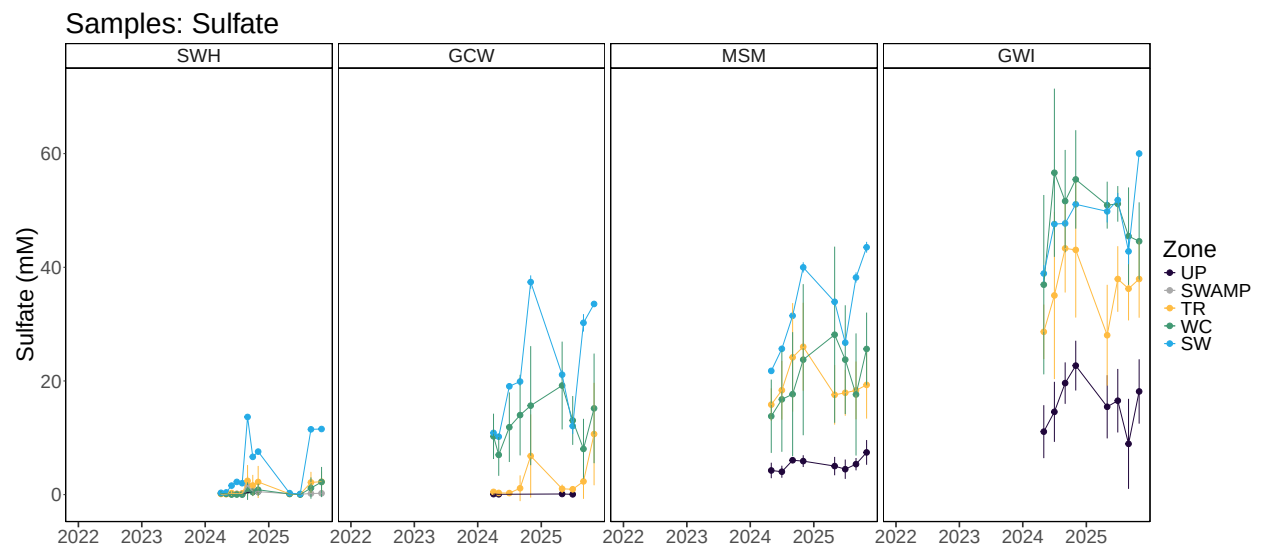
```
## 'summarise()' has grouped output by 'Month', 'Year', 'Site'. You can override
## using the '.groups' argument.

## Warning: Removed 178 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 33 rows containing missing values or values outside the scale range
## ('geom_line()').
```



```
## Warning: Removed 178 rows containing missing values or values outside the scale range
## ('geom_point()').
## Removed 33 rows containing missing values or values outside the scale range
## ('geom_line()').
```

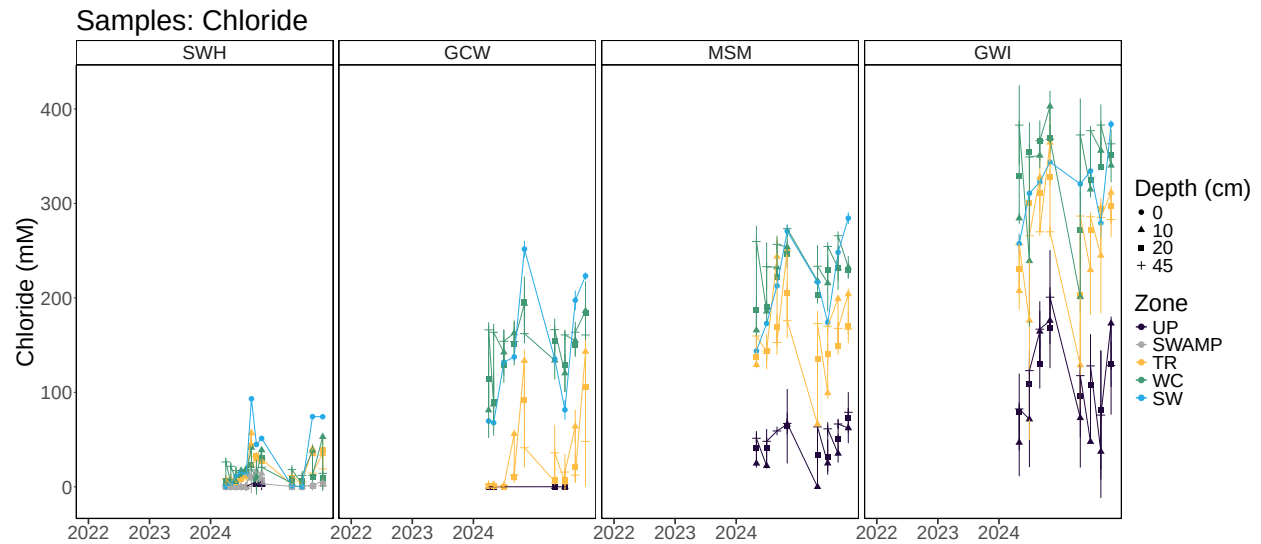


0.5 Summarized data for Depth, Site and Zone

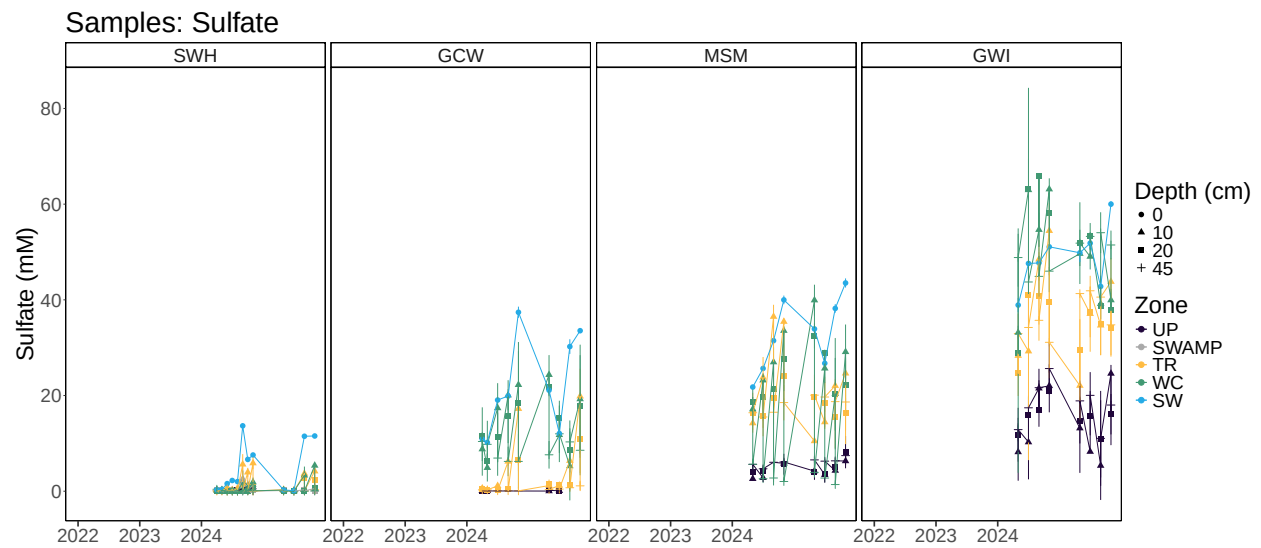
```
## 'summarise()' has grouped output by 'Month', 'Year', 'Site', 'Zone'. You can
## override using the '.groups' argument.
```

```
## Warning: Removed 442 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 87 rows containing missing values or values outside the scale range
## ('geom_line()').
```

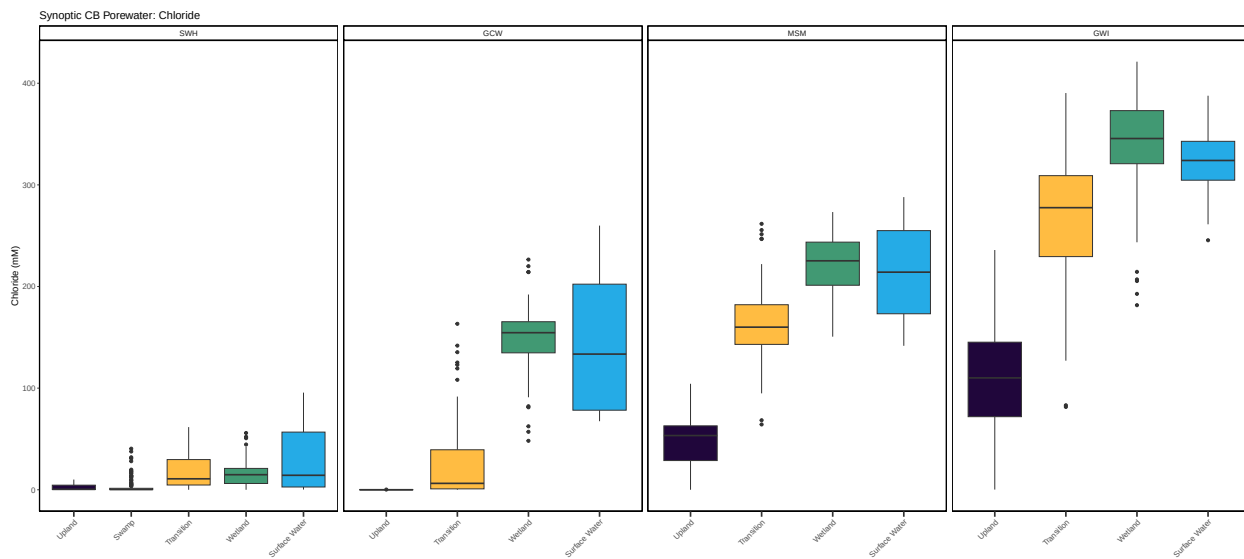


```
## Warning: Removed 442 rows containing missing values or values outside the scale range
## ('geom_point()').
## Removed 87 rows containing missing values or values outside the scale range
## ('geom_line()').
```

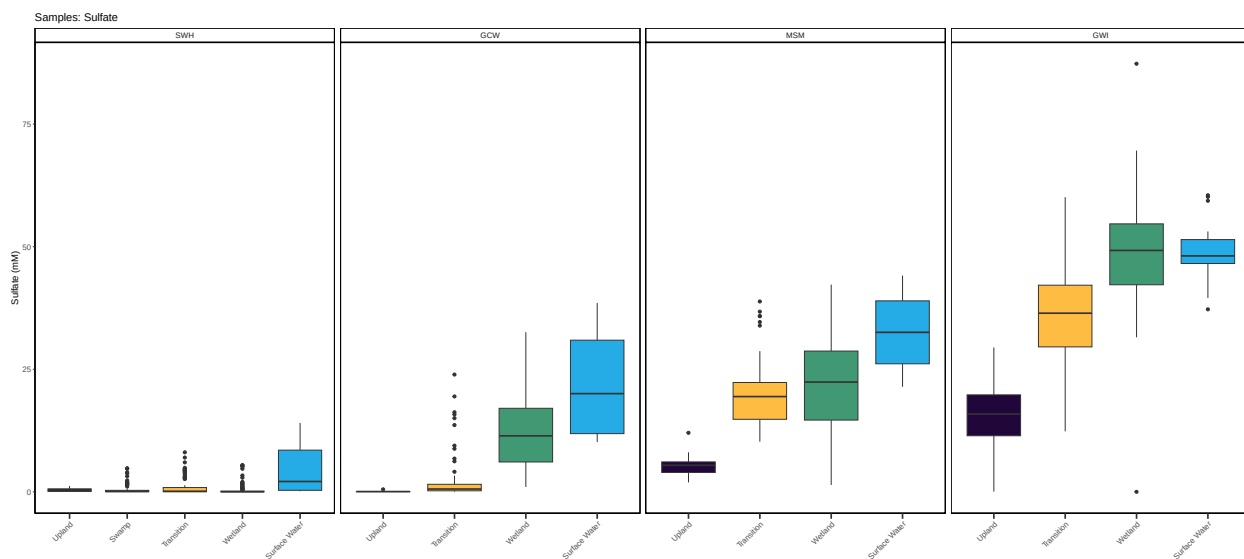


0.6 Boxplot summary

```
## Warning: Removed 1231 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
## Warning: Removed 1231 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
## Warning: Removed 1231 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## pdf
## 2
```

```
## Warning: Removed 1231 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## pdf
## 2
```